Diode, Equivalent Circuits and Zener Diode

Basic Electrical and Electronics Engineering

Unit 3: Diodes, Transistors and their Applications

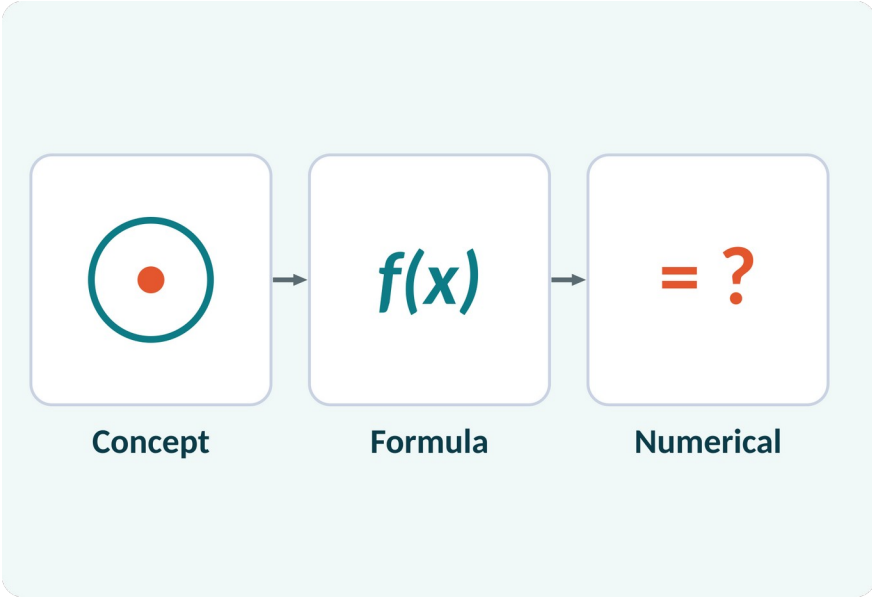
Topic: Semiconductor diode, diode equivalent circuits, Zener diode as voltage regulator

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Learning Outcomes

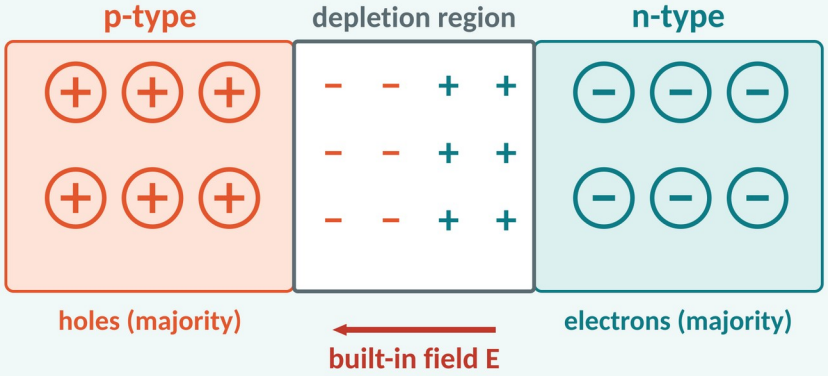
- Explain the p-n junction and its biasing.
- Read the diode V–I characteristic.
- Use ideal and practical diode models.
- Design a simple Zener voltage regulator.



Linked to CO3
Analyse characteristics of the p-n junction diode and Zener diode.

Recap: What You Already Know

- Semiconductors: silicon and germanium.
- Doping makes p-type (holes) and n-type (electrons).
- Majority and minority carriers.
- Ohm's law and resistor circuits.

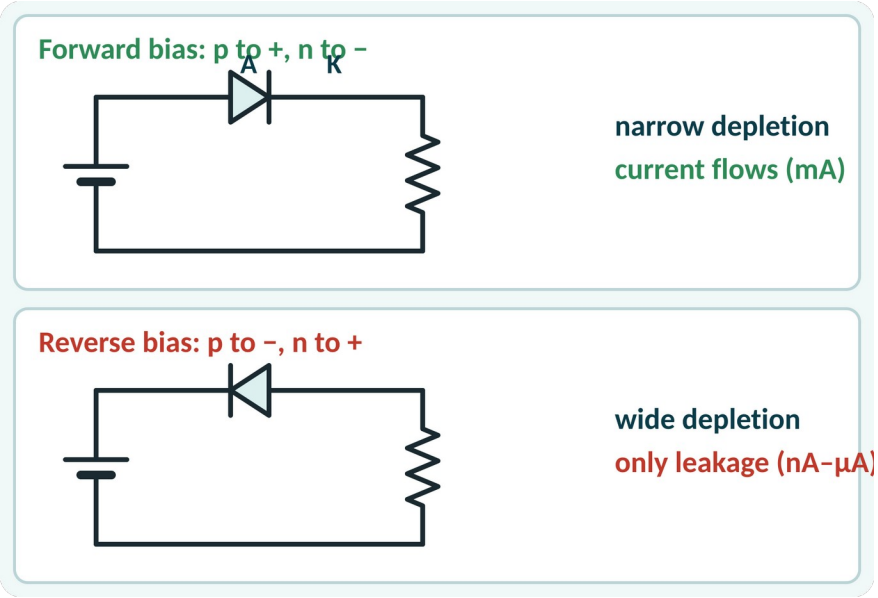


barrier potential: Si ≈ 0.7 V, Ge ≈ 0.3 V

p-type: trivalent dopant (holes); n-type: pentavalent dopant (electrons)

Motivation: A One-Way Valve for Current

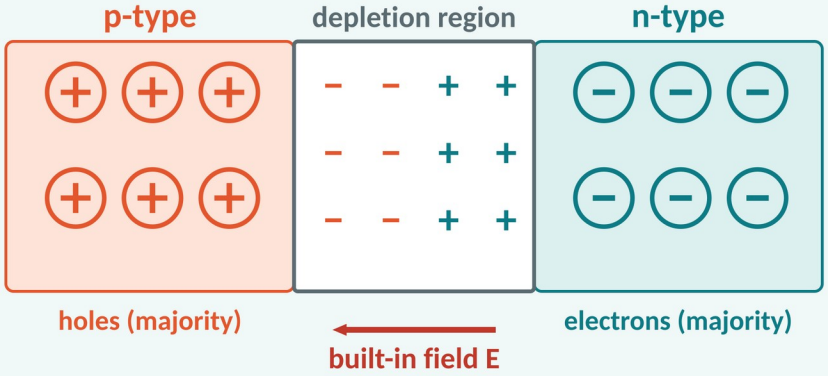
- Mobile charger converts AC to DC with diodes.
- LED lights, radios, solar panels.
- Protects circuits from reverse polarity.
- Zener diodes give a steady reference voltage.



A diode conducts in one direction and blocks in the other

The p-n Junction

- Holes and electrons diffuse across the junction.
- They recombine, leaving fixed ions.
- Depletion region forms; a built-in field appears.
- Barrier potential opposes further diffusion.
- Si ≈ 0.7 V; Ge ≈ 0.3 V.

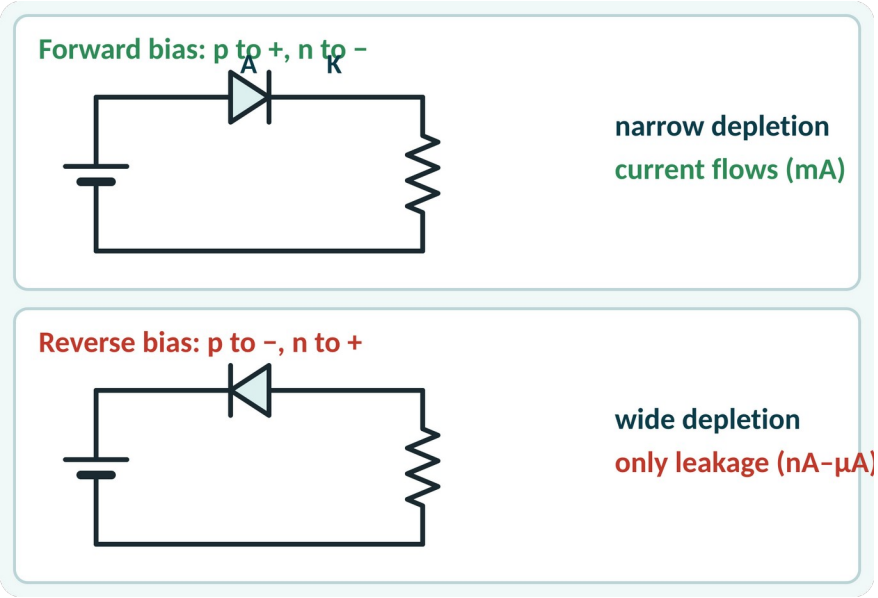


barrier potential: Si ≈ 0.7 V, Ge ≈ 0.3 V

Barrier potential: Si ≈ 0.7 V, Ge ≈ 0.3 V at room temperature

Forward and Reverse Bias

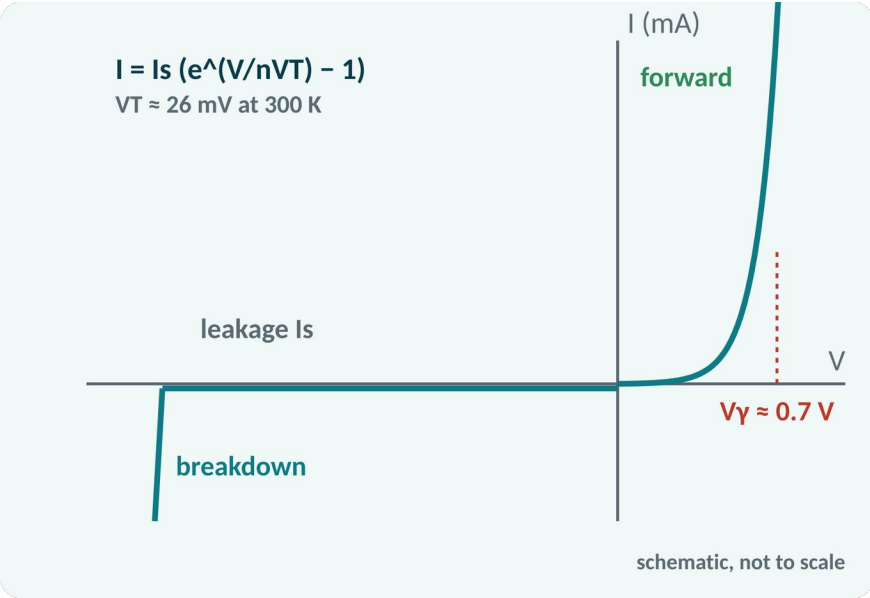
- Forward: p to +, n to − of the source.
- Depletion narrows; current flows above 0.7 V.
- Reverse: p to −, n to +.
- Depletion widens; only a tiny leakage flows.
- Resistor in series limits forward current.



Forward: conducts when V > Vγ; reverse: I ≈ Is (nA–µA) until breakdown

Diode V–I Characteristic

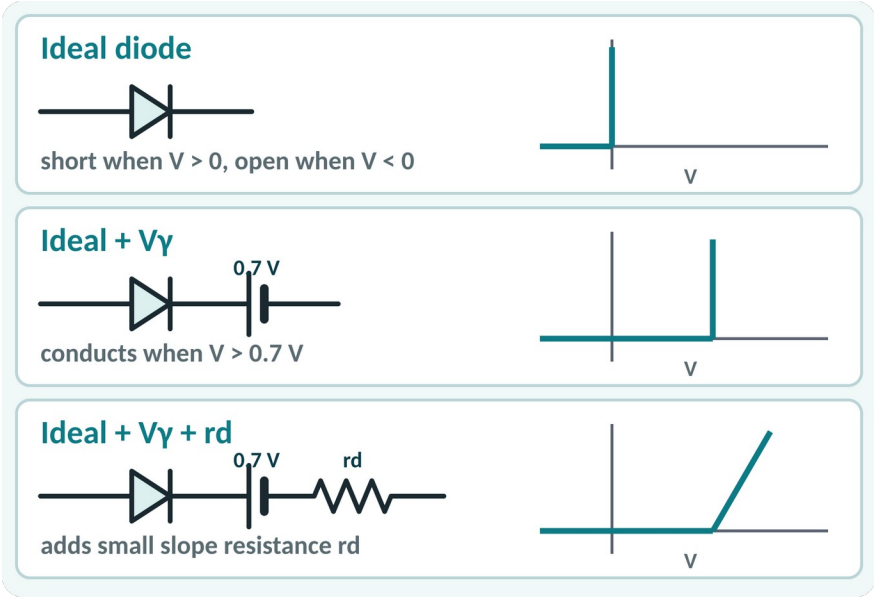
- Forward: exponential rise after the knee.
- Knee voltage Vγ ≈ 0.7 V (Si).
- Reverse: small, nearly constant leakage.
- Breakdown at high reverse voltage.
- Leakage doubles about every 10 °C.



I = Is·(e^(V/(n·VT)) − 1); VT = kT/q ≈ 26 mV at 300 K
Is reverse saturation current, n ideality factor (1–2)

Diode Equivalent Circuits

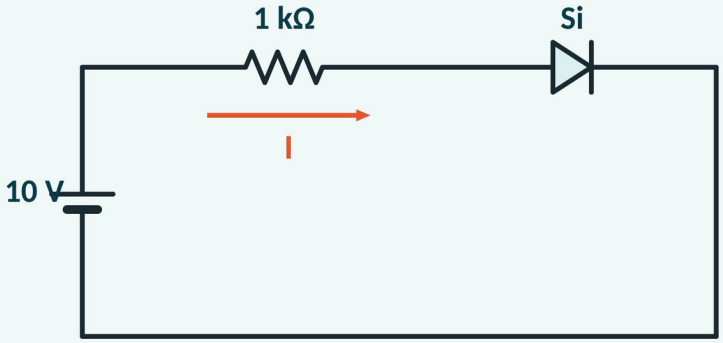
- Ideal: a switch, no drop.
- Practical: ideal diode plus 0.7 V battery.
- Piecewise linear: add a small resistance rd.
- Choose the simplest model that is accurate enough.
- Reverse: open circuit in all models.



Ideal: V = 0 if on; Practical: V = Vγ; Linear: V = Vγ + I·rd

Analysing a Diode Circuit

- Decide: forward or reverse biased?
- Replace the diode by its equivalent circuit.
- Apply KVL around the loop.
- Solve for the current.
- Check the assumption.



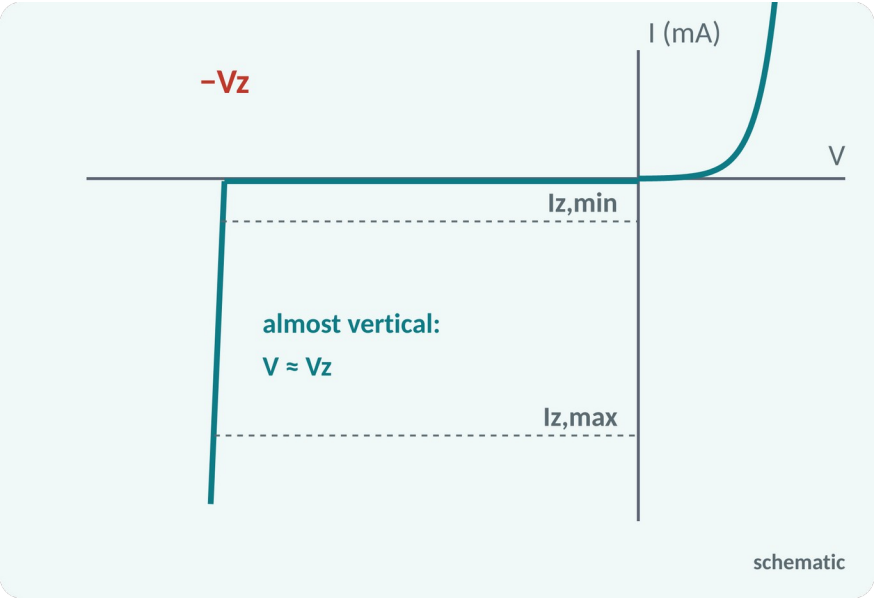
$$I = (V - V_\gamma)/R$$

$$= (10 - 0.7)/1\ k\Omega = 9.3\ mA$$

Series diode–resistor: $I = (V - V_\gamma)/R$ $(V > V_\gamma)$; $I \approx 0$ if reverse biased

Zener Diode

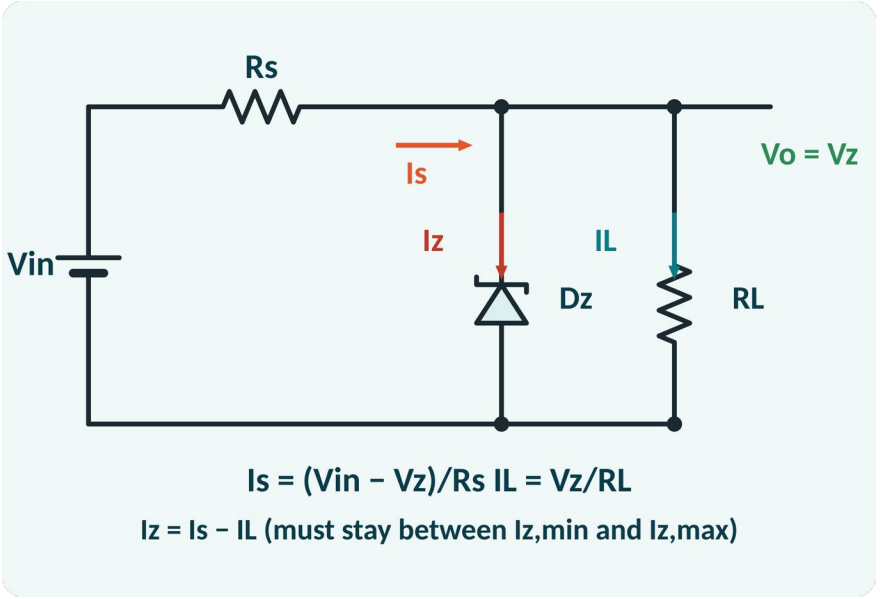
- Heavily doped diode designed to operate in breakdown.
- Reverse breakdown voltage Vz is sharp and stable.
- Voltage stays near Vz over a wide current range.
- Safe current limited by power rating.
- Symbol: diode with bent bar.



Operation: reverse biased, $I_{z,min} < I_z < I_{z,max}$; $P_z = V_z \cdot I_z \le P_{z,max}$

Zener Voltage Regulator

- Series resistor Rs, Zener in parallel with load.
- Output Vo = Vz while the Zener is in breakdown.
- Source current splits: Iz and IL.
- Iz absorbs changes in Vin or load.
- Need Iz,min ≤ Iz ≤ Iz,max.



Is = (Vin − Vz)/Rs; IL = Vz/RL; Iz = Is − IL

Regulator Behaviour and Limits

- Line regulation: Vin changes, Vo steady.
- Load regulation: RL changes, Vo steady.
- Regulation fails when Iz < Iz,min.
- No-load case: Iz = Is must be ≤ Iz,max.
- Choose Rs to meet both limits.

Vin rises (RL fixed)

Is rises, IL same → Iz rises, Vo ≈ Vz

Vin falls (RL fixed)

Is falls → Iz falls; must stay > Iz,min

RL falls (more load)

IL rises → Iz falls; Vo ≈ Vz while Iz > Iz,min

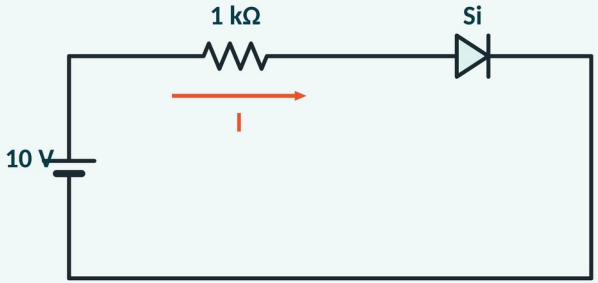
RL open (no load)

all of Is flows in Dz: Iz = Is ≤ Iz,max

$$R_{s,max} = \frac{V_{in,min} - V_z}{I_{z,min} + I_{L,max}}; \quad R_{s,min} = \frac{V_{in,max} - V_z}{I_{z,max} + I_{L,min}}$$

Solved Example 1: Series Diode

- 10 V source, 1 kΩ resistor, silicon diode. Find the current for (a) forward bias with the practical model, (b) ideal diode, (c) reversed diode.
- (a) I = (10 − 0.7)/1 kΩ = 9.3 mA.
- (b) Ideal: I = 10/1 kΩ = 10 mA.
- (c) Reverse: diode off, I ≈ 0, all 10 V across the diode.
- Power in the diode (a): P = 0.7 × 9.3 mA = 6.5 mW.



$$I = (V - V_\gamma)/R$$

$$= (10 - 0.7)/1\ k\Omega = 9.3\ mA$$

Answer: 9.3 mA (practical), 10 mA (ideal), ≈ 0 (reversed)

Solved Example 2: Zener Regulator

- Vin = 20 V, Rs = 200 Ω, Vz = 10 V. Find Is, IL, Iz for RL = 1 kΩ and for RL = 250 Ω. Find RL,min if Iz,min = 5 mA.
- 1. Is = (20 − 10)/200 = 50 mA (both cases).
- 2. RL = 1 kΩ: IL = 10 mA, Iz = 40 mA.
- 3. RL = 250 Ω: IL = 40 mA, Iz = 10 mA.
- 4. IL,max = Is − Iz,min = 45 mA → RL,min = 10/0.045 = 222 Ω.



Answer: Iz = 40 mA and 10 mA; regulation holds for RL ≥ 222 Ω

Common Mistakes and Exam Tips

Common mistakes

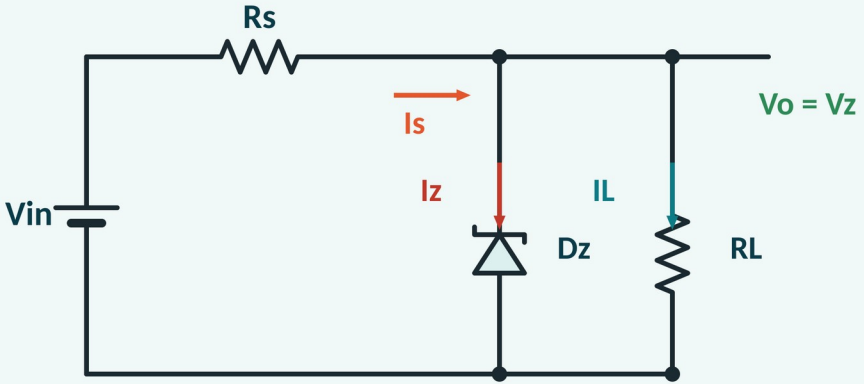
- Using 0.7 V for a Zener in breakdown.
- Connecting the Zener forward-biased.
- No series resistor with the Zener or diode.
- Ignoring the minimum Zener current.
- Assuming the diode is on without checking.

Exam tips

- Check forward or reverse bias first.
- Always include Rs.
- Verify $I_{z,min} \le I_z \le I_{z,max}$.
- Verify the current sign after solving.

Summary

- p-n junction: depletion region, barrier 0.7 V (Si).
- Forward conducts above Vγ; reverse blocks.
- Models: ideal, ideal + Vγ, + rd.
- Zener operates in reverse breakdown at Vz.
- Regulator: Iz = Is − IL stays within limits.



Is = (Vin − Vz)/Rs IL = Vz/RL

Iz = Is − IL (must stay between Iz,min and Iz,max)

I = (V − Vγ)/R; Is = (Vin − Vz)/Rs; Iz = Is − IL

Quick Quiz

Q1. Barrier potential of silicon is about:
(a) 0.3 V (b) 0.7 V (c) 1.2 V (d) 5 V

Ans: (b)

Q2. A diode is forward biased when:
(a) p to −, n to + (b) p to +, n to − (c) both to + (d) open

Ans: (b)

Q3. 10 V, 1 kΩ, Si diode: I =
(a) 10 mA (b) 9.3 mA (c) 0.7 mA (d) 0

Ans: (b)

Q4. Zener diode works in:
(a) forward region (b) reverse breakdown (c) cut-off only (d) saturation

Ans: (b)

Q5. Zener regulator: Iz equals:
(a) Is + IL (b) Is − IL (c) IL − Is (d) Vz/Rs

Ans: (b)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: rectifiers: half-wave and full-wave bridge